Real or nominal? Reading a country's growth figures

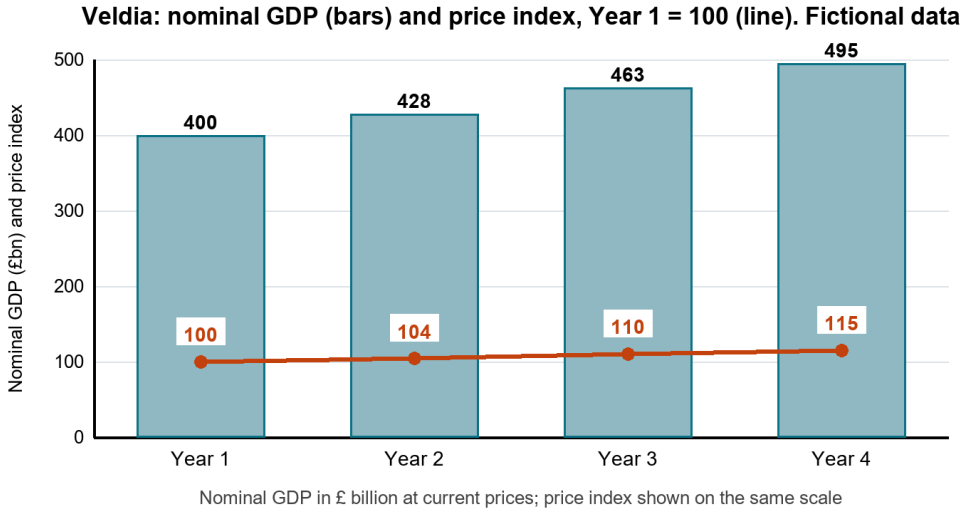
Student tasks | Original TBS lesson for A level Economics (Edexcel 9EC0 content, Theme 2, measures of economic performance)

A government minister in Veldia says, "Our economy grew by 8.2% last year." Use the data below to find out what that figure really measures, whether people are better off, and what the minister should have said.

Before you start you should know: nominal GDP is measured at the prices of the year it was produced; real GDP is measured at the prices of one fixed year, so it removes the effect of price changes. Real GDP = nominal GDP / price index x 100. Percentage change = (new - old) / old x 100. You need a calculator. This is original classroom practice, not an exam paper. Veldia and every figure are fictional.

The data

Veldia	Year 1	Year 2	Year 3	Year 4
Nominal GDP (£ billion)	400	428	463	495
Price index (Year 1 = 100)	100	104	110	115
Population (millions)	20.0	20.2	20.4	20.6



Round real GDP to one decimal place and percentage changes to one decimal place. Use the rounded real GDP figures for later steps.

Starter: which measure?

Write N (nominal) or R (real), and T (total) or P (per person), for each description.

Description	N or R	T or P
(i) GDP measured at the prices of the year it was produced		
(ii) GDP measured at constant prices of a base year		
(iii) GDP divided by the population		
(iv) the figure whose rate of change is normally used as the measure of economic growth		

Task 1: real GDP

Complete the table. Year 1 is the base year, so real GDP is the same as nominal GDP in Year 1.

£ billion	Year 1	Year 2	Year 3	Year 4
Real GDP (Year 1 prices)	400.0			

Show one line of working for Year 2.

Task 2: growth and inflation

Calculate each percentage change from the previous year.

% change on previous year	Year 2	Year 3	Year 4
Nominal GDP			
Real GDP			
Price index (inflation)			

Working space:

a. In each year, which is larger, nominal growth or real growth? Explain the gap using the price index.

b. Which rate is the better measure of economic growth, and why?

Task 3: evaluate the minister's claim

"Our economy grew by 8.2% last year." Assume the minister is talking about Year 3.

a. Which figure from your table is the 8.2%, and is the claim misleading? Explain.

b. Write one sentence the minister could say instead, using the figures.

Task 4: living standards

a. Calculate real GDP per person for Year 1 and Year 4, in £ per person to the nearest £1 (real GDP in £ billion divided by population in millions, multiplied by 1,000). Then calculate the percentage change.

	Year 1	Year 4	% change
Real GDP per person (£)			

b. Total real GDP grew by 7.6% between Year 1 and Year 4. Why is the per person growth lower?

Task 5: decide

Did living standards in Veldia improve between Year 1 and Year 4? Write about 100 words. Use at least three figures from your tables, and explain one limitation of using GDP to compare living standards over time.

Exit ticket

Country Z: nominal GDP rises from £250 billion to £270 billion and its price index rises from 100 to 106.

a. Calculate nominal growth, real GDP in the new year (to one decimal place, in the old prices) and real growth.

b. In one sentence, explain why real growth is so much lower than nominal growth.